EXERCISE 13

# Multi-Equilibria Autonomous Phase Line Dynamics

Original statement · ProblemsCh2.pdf, p. 3

Phase line • Finite-time blow-up

PROBLEM STATEMENT

Consider the autonomous differential equation

$$\frac{dy}{dt} = y(y-2)(y-3).$$

In each of the following cases, what can you conclude based on the uniqueness theorem about the different solutions if

(i) $y(0) = -1$,
(ii) $y(0) = 1$,
(iii) $y(0) = 2$,
(iv) $y(0) = 4$?

Source: ProblemsCh2.pdf, Exercise 13 (page 3).

PHASE 1: CLASSIFICATION, HYPOTHESES AND DOMAIN

Autonomous equation $\dot{y} = f(y)$ with the cubic polynomial

$$f(y) = y(y-2)(y-3) = y^3 - 5y^2 + 6y, \qquad f'(y) = 3y^2 - 10y + 6.$$

Both $f$ and $f'$ are continuous on all of $\mathbb{R}$, so the existence and uniqueness theorem holds at every initial point: through each $(0, y_0)$ passes exactly one solution, defined on a maximal open interval of existence.

Equilibria ($f(y^*) = 0$): $y^* = 0,\ 2,\ 3$. They give the constant solutions $y \equiv 0, y \equiv 2, y \equiv 3$.

PHASE 2: CHOICE OF METHOD AND CHANGE OF VARIABLES

Why this method: by uniqueness, two different solutions of the same autonomous equation never touch. A non-constant solution therefore cannot reach an equilibrium value in finite time, so it stays in one of the four intervals

$$I_1 = (-\infty, 0), \quad I_2 = (0, 2), \quad I_3 = (2, 3), \quad I_4 = (3, \infty)$$

determined by the equilibria. Inside an interval $f$ has a constant sign, so $y(t)$ is strictly monotone. A monotone function that stays in a bounded interval has a limit, and this limit must be an equilibrium. If the interval is unbounded, the solution may escape to infinity; whether this happens in finite time is decided by separating variables.

Sign of $f$ and stability from $f'$:

| **Interval or point** | **Sign of $f$ (test point)** | **Motion** |
|---|---|---|
| $y < 0$ | $f(-1) = (-1)(-3)(-4) = -12 < 0$ | decreasing |
| $0 < y < 2$ | $f(1) = (1)(-1)(-2) = 2 > 0$ | increasing |
| $2 < y < 3$ | $f(2.5) = (2.5)(0.5)(-0.5) = -0.625 < 0$ | decreasing |
| $y > 3$ | $f(4) = (4)(2)(1) = 8 > 0$ | increasing |

Stability: $f'(0) = 6 > 0$ (unstable), $f'(2) = 12 - 20 + 6 = -2 < 0$ (asymptotically stable), $f'(3) = 27 - 30 + 6 = 3 > 0$ (unstable). This agrees with the arrows: flow leaves $0$ and $3$ on both sides and enters $2$ from both sides.

Time scale for blow-up: for $|y|$ large, $f(y) \sim y^3$, and the time needed to reach infinity is $\int^{\infty} dy/f(y) \sim \int^{\infty} dy/y^3 < \infty$. This finite integral signals that infinity is reached in finite time. We compute the time exactly by separation of variables.

METHOD OF THE CLASS NOTES [NOTES, P. 3]

The notes classify a stationary point $x^*$ ($f(x^*) = 0$) by the sign of $f'(x^*)$: $f'(x^*) > 0$ unstable, $f'(x^*) < 0$ stable. With $f(y) = y(y-2)(y-3) = y^3 - 5y^2 + 6y$, $f'(y) = 3y^2 - 10y + 6$, which gives the three values above. The sketch argument of the notes (the solution moves right where $f > 0$ and left where $f < 0$) is the sign table, and both routes give the same classification.

PHASE 3: STEP-BY-STEP DERIVATION

### Separation of variables with partial fractions

For $y \neq 0, 2, 3$ write $\dfrac{dy}{y(y-2)(y-3)} = dt$ and decompose:

$$\frac{1}{y(y-2)(y-3)} = \frac{A}{y} + \frac{B}{y-2} + \frac{C}{y-3}.$$

Cover-up rule: $A = \frac{1}{(0-2)(0-3)} = \frac{1}{6}$, $B = \frac{1}{2\,(2-3)} = -\frac{1}{2}$, $C = \frac{1}{3\,(3-2)} = \frac{1}{3}$. Check: $A + B + C = \frac{1}{6} - \frac{1}{2} + \frac{1}{3} = 0$, as required since the left side decays like $y^{-3}$. Integrating,

$$G(y) := \int \frac{dy}{y(y-2)(y-3)} = \frac{1}{6}\ln|y| - \frac{1}{2}\ln|y-2| + \frac{1}{3}\ln|y-3|.$$

Barrow's rule from $(0, y_0)$: $G(y(t)) - G(y_0) = t$. Equivalently, multiplying by 6 and exponentiating,

$$\frac{|y|\,|y-3|^2}{|y-2|^3} = K\,e^{6t}, \qquad K = \frac{|y_0|\,|y_0-3|^2}{|y_0-2|^3}.$$

Limit of $G$ at infinity: the coefficients sum to zero, so $G(y) = \frac{1}{6}\ln\dfrac{|y|\,|y-3|^2}{|y-2|^3}$, and the argument of the logarithm tends to 1 as $y \to \pm\infty$. Hence $G(\pm\infty) = \frac{1}{6}\ln 1 = 0$.

**Case (i):** $y(0) = -1$

$y(0) \in I_1$. By uniqueness the solution cannot reach the equilibrium $y = 0$, so $y(t) < 0$ for all $t$ in its domain. Since $f < 0$ there, $y$ is strictly decreasing.

- Backward in time: $y$ increases and is bounded above by 0, so it tends to an equilibrium: $y \to 0^-$ as $t \to -\infty$.
- Forward in time: $y$ decreases without a lower barrier and $f \sim y^3$, so it blows up to $-\infty$ at a finite time $T^*$.

Blow-up time from Barrow's rule, $T^* = G(-\infty) - G(-1) = -G(-1)$:

$$G(-1) = \frac{1}{6}\ln 1 - \frac{1}{2}\ln 3 + \frac{1}{3}\ln 4, \qquad T^* = \frac{1}{2}\ln 3 - \frac{1}{3}\ln 4 = \frac{1}{2}\ln 3 - \frac{2}{3}\ln 2 \approx 0.0872.$$

Cross-check with the closed form: $K = \frac{1\cdot 16}{27}$, and blow-up occurs when the left side tends to 1 (as $|y| \to \infty$): $1 = \frac{16}{27}e^{6T^*}$, so $T^* = \frac{1}{6}\ln\frac{27}{16} = \frac{1}{6}(3\ln 3 - 4\ln 2)$, the same value.

**Case (ii):** $y(0) = 1$

$y(0) \in I_2 = (0, 2)$. The equilibria 0 and 2 are barriers, so $0 < y(t) < 2$ for all $t$. A bounded solution cannot blow up, so it is defined for all $t \in \mathbb{R}$. Since $f > 0$, $y$ is strictly increasing, hence it has limits at both ends and they are equilibria:

$$\lim_{t\to-\infty} y(t) = 0, \qquad \lim_{t\to+\infty} y(t) = 2.$$

The solution goes from the unstable equilibrium $0$ to the stable equilibrium $2$ (a monotone, sigmoidal profile). Neither limit is reached in finite time.

**Case (iii):** $y(0) = 2$

$f(2) = 0$, so $y(t) \equiv 2$ is a solution of the IVP. By the uniqueness theorem it is the only one:

$$y(t) \equiv 2 \quad \forall t \in \mathbb{R}.$$

In particular no non-constant solution can pass through the point $(t, 2)$ at any time; solutions that start in $I_2$ or $I_3$ approach $2$ only asymptotically.

**Case (iv):** $y(0) = 4$

$y(0) \in I_4 = (3, \infty)$. The barrier $y = 3$ gives $y(t) > 3$ for all $t$; $f > 0$ there, so $y$ is strictly increasing.

- Backward in time: bounded below by $3$ and decreasing as $t$ decreases, so $y \to 3^+$ as $t \to -\infty$.
- Forward in time: no upper barrier and $f \sim y^3$, so $y \to +\infty$ at a finite time $T^*$.

$$T^* = G(+\infty) - G(4) = -G(4) = -\left[\frac{1}{6}\ln 4 - \frac{1}{2}\ln 2 + \frac{1}{3}\ln 1\right] = -\left[\frac{1}{3}\ln 2 - \frac{1}{2}\ln 2\right] = \frac{1}{6}\ln 2 \approx 0.1155.$$

Cross-check with the closed form: $K = \frac{4\cdot 1}{8} = \frac{1}{2}$, and $1 = \frac{1}{2}e^{6T^*}$ gives $T^* = \frac{1}{6}\ln 2$.

PHASE 4: VERIFICATION, LIMITS AND INTERPRETATION

Numerical checks (computed): the partial-fraction decomposition from sympy is $\frac{1}{6y} - \frac{1}{2(y-2)} + \frac{1}{3(y-3)}$, matching $A, B, C$. An independent fourth-order Runge-Kutta integration of $\dot y = y(y-2)(y-3)$ with step $1.5 \times 10^{-6}$ reaches $|y| > 10^7$ at $t \approx 0.115526$ for $y_0 = 4$ (formula: $0.115525$) and at $t \approx 0.087210$ for $y_0 = -1$ (formula: $0.087208$).

Time scale: the correct statement for $y_0 = -1$ and $y_0 = 4$ is not $y \to \pm\infty$ as $t \to \infty$; the solutions cease to exist at $T^* \approx 0.087$ and $T^* \approx 0.116$ respectively, so a limit $t \to \infty$ is meaningless for them. By contrast, for $y_0 = 1$ the solution is global and the limit $t \to \infty$ is $2$.

Dimensional remark: if $y$ is a non-dimensional state and $t$ is measured in units of a characteristic time $\tau$, then $T^*$ is a number of such units; the equation $\dot y = f(y)$ requires $[f] = [y]/[t]$, and the coefficients $2, 3$ carry the dimensions of $y$.

CLASSIFICATION OF TRAJECTORIES

| Initial condition | Region | Behavior forward | Behavior backward |
|---|---|---|---|
| (i) $y(0) = -1$ | $y < 0$, decreasing | blow-up to $-\infty$ at $T^* = \frac{1}{2}\ln 3 - \frac{2}{3}\ln 2 \approx 0.087$ | $y \to 0^-$ as $t \to -\infty$ |
| (ii) $y(0) = 1$ | $0 < y < 2$, increasing | $y \to 2$ as $t \to \infty$ | $y \to 0^+$ as $t \to -\infty$ |
| (iii) $y(0) = 2$ | equilibrium | $y \equiv 2$ | $y \equiv 2$ |
| (iv) $y(0) = 4$ | $y > 3$, increasing | blow-up to $+\infty$ at $T^* = \frac{1}{6}\ln 2 \approx 0.116$ | $y \to 3^+$ as $t \to -\infty$ |